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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (E&E/E&C) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EE301/20EC301/17EE301/21EC301/20EC301- VECTOR CALCULUS AND TRANSFORM TECHNIQUES

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1, -1, 1)$ in the direction of the vector $2\hat{i} - \hat{j} - 2\hat{k}$.	6	L*1	1	2
b)	Prove that the cylindrical polar co-ordinate system is an orthogonal curvilinear co-ordinate system.	7	L2	1	1
c)	Using Green's theorem, evaluate $\int_C (xy^2 - 2xy)dx + (x^2y + 3)dy$ around the boundary of the region enclosed by $y^2 = 8x$ and $x = 2$.	7	L2	2	2
2. a)	Show that the vector field $\vec{F} = (2xy + z^3)\hat{i} + x^2\hat{j} + 3xz^2\hat{k}$ is conservative. Also find its scalar potential.	6	L2	2	2
b)	With usual notation prove that $\int_C \vec{F} \cdot d\vec{r}$ is independent of the path of integration if and only if $\oint_C \vec{F} \cdot d\vec{r} = 0$. Where C is a simple closed curve.	7	L2	2	1
c)	A particle moves along the curve $x = 2t^2, y = t^2 - 4t, z = -t - 5$, where t is the time variable. Find the components of its velocity and acceleration at time $t = 1$ in the direction given by $\hat{i} - 2\hat{j} + 2\hat{k}$.	7	L1	1	2
3. a)	Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ & $z = x^2 + y^2 - 3$ at $(1, 2, 2)$.	6	L1	1	2
b)	Using Gauss divergence theorem, evaluate $\iiint_V \vec{F} \cdot \hat{n} dS$, where $\vec{F} = (y - z + 2)\hat{i} - (yz + 4)\hat{j} - xz\hat{k}$ and S is the closed surface of the cube bounded by $x = 0, x = 2, y = 0, y = 2, z = 0$ & $z = 2$.	7	L2	2	2
c)	State and prove Green's theorem in the plane.	7	L2	2	1
Unit – II					
4. a)	Find the bilinear transformation which maps $z = 1, i, -1$ onto $w = 0, 1, \infty$.	6	L1	3	2
b)	If $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D , then prove that at any point of D , $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$.	7	L2	3	1
c)	State Cauchy's residue theorem. Using it evaluate $\int_C \frac{12z-7}{(z-1)^2(2z+3)} dz$, where C is the circle $ z = 2$.	7	L2	4	2
5. a)	State and prove Cauchy's integral theorem.	6	L2	4	1
b)	Using residue theorem evaluate $\int_C \frac{2z-1}{z(z+1)(z-3)} dz$, where C is the circle $ z = 2$.	7	L2	4	2
c)	If $f(z)$ is a regular function of z , prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) Re f(z) ^2 = 2 f'(z) ^2$.	7	L1	3	2

P.T.O.

6. a) Determine the analytic function $f(z)$ whose real part is $u = y + e^x \cos y$. Also find $f'(z)$. 6 L1 3 2
- b) Obtain the Laurent's expansion of $f(z) = \frac{1}{(z+1)(z+3)}$ in the region (i) $1 < |z| < 3$ (ii) $|z| > 3$. 7 L2 4 2
- c) State and prove Cauchy's integral formula. 7 L2 4 1

Unit – III

7. a) Find the half range Fourier cosine series of $f(x) = x, 0 < x < 2$. 6 L1 5 2
- b) Obtain the Fourier series of $f(x) = \begin{cases} 1-x & , -\pi \leq x \leq 0 \\ 1+x & , 0 \leq x \leq \pi \end{cases}$ and $f(x+2\pi) = f(x)$. 7 L3 5 2
- c) Find the Fourier series containing the first harmonics from the following table.
- | | | | | | | |
|-----------|-----|-----|-----|-----|-----|-----|
| x° | 0 | 60 | 120 | 180 | 240 | 300 |
| y | 7.9 | 7.2 | 3.6 | 0.5 | 0.9 | 6.8 |
- 7 L1 5 2
8. a) Find the bilateral Z-transform and region of convergence of $u_n = \begin{cases} 5^n & \text{if } n < 0 \\ 3^n & \text{if } n \geq 0 \end{cases}$. 6 L1 5 2
- b) State and prove convolution theorem for bilateral Z-transforms. 7 L1 5 1
- c) Using unilateral Z-transforms, solve $u_{n+2} + 4u_{n+1} + 3u_n = 3^n$ with $u_0 = 0, u_1 = 1$. 7 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
